

Research paper

Eccentric rehabilitation induces white matter plasticity and sensorimotor recovery in chronic spinal cord injury

Timothy D. Faw^{a,b,c}, Bimal Lakhani^d, Petra Schmalbrock^e, Michael V. Knopp^e,
Keith R. Lohse^{f,g}, John L.K. Kramer^{h,i}, Hanwen Liu^{i,j}, Huyen T. Nguyen^e, Eileen G. Phillips^{b,k},
Anna Bratasz^l, Lesley C. Fisher^{b,k}, Rochelle J. Deibert^{b,k}, Lara A. Boyd^d, Dana M. McTigue^{a,b,m},
D. Michele Basso^{a,b,k,*}

^a Neuroscience Graduate Program, The Ohio State University, Columbus, OH 43210, USA

^b Center for Brain and Spinal Cord Repair, The Ohio State University, Columbus, OH 43210, USA

^c Department of Orthopaedic Surgery, Duke University, Durham, NC 27710, USA

^d Department of Physical Therapy, University of British Columbia, Vancouver, BC V6T 1Z3, Canada

^e Department of Radiology, The Ohio State University, Columbus, OH 43210, USA

^f Department of Health, Kinesiology, and Recreation, University of Utah, Salt Lake City, UT 84112, USA

^g Department of Physical Therapy and Athletic Training, University of Utah, Salt Lake City, UT 84108, USA

^h Department of Anesthesiology, Pharmacology, and Therapeutics, University of British Columbia, Vancouver, BC V6T 1Z3, Canada

ⁱ International Collaboration on Repair Discoveries, University of British Columbia, Vancouver, BC V5Z 1M9, Canada

^j Department of Physics and Astronomy, University of British Columbia, Vancouver, BC V6T 1Z1, Canada

^k School of Health and Rehabilitation Sciences, The Ohio State University, Columbus, OH 43210, USA

^l Small Animal Imaging Shared Resources, Davis Heart and Lung Research Institute, The Ohio State University, Columbus, OH 43210, USA

^m Department of Neuroscience, The Ohio State University, Columbus, OH 43210, USA



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ABSTRACT

Experience-dependent white matter plasticity offers new potential for rehabilitation-induced recovery after neurotrauma. This first-in-human translational experiment combined myelin water imaging in humans and genetic fate-mapping of oligodendrocyte lineage cells in mice to investigate whether downhill locomotor rehabilitation that emphasizes eccentric muscle actions promotes white matter plasticity and recovery in chronic, incomplete spinal cord injury (SCI). In humans, of 20 individuals with SCI that enrolled, four passed the imaging screen and had myelin water imaging before and after a 12-week (3 times/week) downhill locomotor treadmill training program (SCI + DH). One individual was excluded for imaging artifacts. Uninjured control participants ($n = 7$) had two myelin water imaging sessions within the same day. Changes in myelin water fraction (MWF), a histopathologically-validated myelin biomarker, were analyzed in *a priori* motor learning and non-motor learning brain regions and the cervical spinal cord using statistical approaches appropriate for small sample sizes. *PDGFRα-CreER^{T2};mT/mG* mice, that express green fluorescent protein on oligodendrocyte precursor cells and subsequent newly-differentiated oligodendrocytes upon tamoxifen-induced recombination, were either naive ($n = 6$) or received a moderate (75 kilodyne), contusive SCI at T9 and were randomized to downhill training ($n = 6$) or unexercised groups ($n = 6$). We initiated recombination 29 days post-injury, seven days prior to downhill training. Mice underwent two weeks of daily downhill training on the same 10% decline grade used in humans. Between-group comparison of functional (motor and sensory) and histological (oligodendrogenesis, oligodendroglial/axon interaction, paranodal structure) outcomes occurred post-training. In humans with SCI, downhill training increased MWF in brain motor learning regions (postcentral, precuneus) and mixed motor and sensory tracts of the ventral cervical spinal cord compared to control participants ($P < 0.05$). In mice with thoracic SCI, downhill training induced oligodendrogenesis in cervical dorsal and lateral white matter, increased axon-oligodendroglial interactions, and normalized paranodal structure in dorsal column sensory tracts ($P < 0.05$). Downhill training improved sensorimotor recovery in mice by normalizing hip and knee motor control and reducing hyperalgesia, both of which were associated with new oligodendrocytes in the cervical dorsal columns ($P < 0.05$). Our findings indicate that eccentric-focused, downhill rehabilitation promotes white matter plasticity

* Corresponding author at: 106A Atwell Hall, 453 W. 10th Ave, Columbus, OH 43210, USA.

E-mail address: michele.basso@osumc.edu (D.M. Basso).

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and improved function in chronic SCI, likely *via* oligodendrogenesis in nervous system regions activated by the training paradigm. Together, these data reveal an exciting role for eccentric training in white matter plasticity and sensorimotor recovery after SCI.

1. Introduction

White matter plasticity is a novel mechanism by which the central nervous system (CNS) can change with experience (Borich et al., 2014; Hofstetter et al., 2013; Lakhani et al., 2016; McKenzie et al., 2014; Sampaio-Baptista et al., 2013; Scholz et al., 2009; Xiao et al., 2016), offering new potential for rehabilitation-induced recovery after neurotrauma. Neurons repeatedly activated during motor skill training release neurotransmitters, not only at the synapse but also along the length of the axon (Kukley et al., 2007; Ziskin et al., 2007). These neurotransmitters bind to receptors on oligodendrocyte progenitor cells (OPCs) causing their proliferation and differentiation into mature, myelinating oligodendrocytes (Barres and Raff, 1993, 1999; Bergles et al., 2000; Borges et al., 1994; Demerens et al., 1996; Gallo et al., 2008; Gibson et al., 2014; Hamilton et al., 2009; Kukley et al., 2007; Lin and Bergles, 2002, 2004; Malone et al., 2013; Stevens et al., 2002; Stevens et al., 1998; Ziskin et al., 2007). In fact, the molecular programs that lead to new oligodendrocytes and myelin formation are activated within hours of motor learning (Xiao et al., 2016) and are ultimately required for motor learning to occur (McKenzie et al., 2014). Importantly, even small changes in myelination can greatly impact neural network function (Fields et al., 2015; Pajevic et al., 2014), which is especially relevant in spinal cord injury (SCI) where myelin damage and sensorimotor dysfunction are hallmark features.

Mechanical trauma and secondary injury cascades from SCI lead to oligodendrocyte death over weeks, months, or even years after injury in larger species (Almad et al., 2011; Casha et al., 2001; Crowe et al., 1997; Grossman et al., 2001; Hesp et al., 2015; Li et al., 1996; Li et al., 1999; Liu et al., 1997; Shuman et al., 1997; Warden et al., 2001). In mice, injury-induced remyelination by OPCs begins two weeks after SCI and peaks around the lesion by five weeks (Pukos et al., 2019). While the contribution of spontaneous oligodendrogenesis and remyelination to motor recovery remains unsettled, rehabilitation appears to have the potential to promote white matter plasticity. Indeed, skill training after pharmacologic demyelination improves function by stimulating remyelination by new and surviving oligodendrocytes, with the predominant and most effective response by new oligodendrocytes (Bacmeister et al., 2020; Neely et al., 2020). Whether rehabilitation can effectively drive white matter plasticity, motor learning, and impact functional recovery in preclinical or clinical SCI is unknown.

Locomotor deficits persist after preclinical and clinical SCI even following intensive traditional rehabilitation (Buehner et al., 2012; Faw et al., 2018; Hansen et al., 2013; Hansen et al., 2012), warranting novel interventions to maximize plasticity and recovery. Downhill treadmill locomotion is a novel approach that induces immediate neuromotor control benefits in rats with SCI (Hansen et al., 2012); whether these translate to sustained functional improvement is unknown. Downhill locomotion relies almost entirely on eccentric, or lengthening, muscle actions (Lay, 2005; Lay et al., 2006; Lay et al., 2007; Minetti et al., 2002) that uniquely activate the CNS and require spinal integration of afferent signals with descending motor commands (Duchateau and Enoka, 2016). After SCI, such descending and afferent systems appear seminal for recovery of effective locomotion (Akay et al., 2014; Bonizzato et al., 2018; DiGiovanna et al., 2016; Grillner and Dubuc, 1988; Grillner and Zanger, 1984; Isakov et al., 2016; Rossignol et al., 1988; Takeoka and Arber, 2019; Takeoka et al., 2014). Moreover, because oligodendrogenesis results from complex activation of the CNS (McKenzie et al., 2014), downhill training may be especially suited to promote white matter plasticity after SCI.

Here, we combine myelin water imaging in humans with genetic

fate-mapping of oligodendrocyte lineage cells in mice to determine whether complex, downhill training at a chronic timepoint induces white matter plasticity and functional recovery in preclinical and clinical SCI. We hypothesized that training-induced white matter plasticity would occur remote from the injury site, along pathways likely activated by the training paradigm. Myelin water imaging revealed increased myelin water fraction (MWF), an imaging biomarker of white matter, in brain motor learning regions and mixed motor and sensory tracts of the ventral cervical spinal cord after downhill training. In mice, training-induced oligodendrogenesis occurred in dorsal column sensory tracts and was associated with motor and sensory recovery. Together, these findings identify a novel approach that induces white matter plasticity and recovery in chronic SCI.

2. Materials and methods

2.1. Experimental procedures in human participants

The myelin imaging study recruited participants from two ongoing clinical trials designed to investigate the effect of eccentric-focused, downhill training on motor control after SCI (NCT02498548, NCT02821845). All individuals provided informed consent in accordance with the Declaration of Helsinki and all aspects of the study were approved by The Ohio State University's (OSU) Institutional Review Board.

2.2. Experimental procedures in human participants

Participants with SCI ($n = 20$) were recruited from the primary study and met the following inclusion criteria: age 18–85 years old, motor incomplete SCI within C1–T10, some capacity to step over ground and on a treadmill, had completed an intensive treadmill-based locomotor training program, and were at least 6 months since discharge from outpatient rehabilitation. Exclusion criteria were: lower motor neuron injury innervating the legs, co-existing neurological conditions (*i.e.*, brain injury), cognitive conditions that prevented informed consent, pregnancy, active deep vein thrombosis, skin wounds in regions where harness or hands provide support, or a history of Botox use for spasticity management within the 3 months prior to enrollment. Age-matched, uninjured control participants ($n = 7$) were recruited from the primary study and the community. All participants completed MRI safety screening. The disposition of participants and reasons for exclusion is shown in Supplementary Fig. 1. See Supplementary Table 1 for demographics of participants who completed the study and all data processing.

2.3. Rehabilitation paradigm

Downhill training for human participants with SCI began at least 6 months after neurorehabilitation ended (Fig. 1). Trainer-assisted downhill stepping occurred 3 times a week for 12 weeks on a 10% grade, using an instrumented, split-belt treadmill (Bertec; Columbus, OH) with custom body-weight support system. Training duration totaled 20 min per session, performed in up to 5-min stepping bouts separated by at least three, 5-min seated rests to prevent delayed onset muscle soreness. Individualized eccentric training complexity occurred by modulating body-weight support and treadmill walking speed to target frontal plane hip motion or sagittal plane knee motion (Worthen-Chaudhari et al., 2015). A team led by a licensed physical therapist in collaboration with up to three clinical rehabilitation assistants provided

intermittent manual assistance to allow error-based motor learning. The objective was self-correction by participants with SCI to attain normal kinematic alignment of the head, trunk, pelvis and legs. Trainers intervened briefly when biomechanical stepping errors predominated. Verbal cues during stepping bouts promoted knowledge of performance, and verbal feedback during rest breaks provided knowledge of results. Verbal cues were structured to facilitate external focus of attention when possible.

2.4. MRI acquisition

Image acquisition occurred at the OSU Wright Center for Innovation using a 3T MRI system (Ingenia CX, Philips Healthcare). To acquire brain images, a dStream head/spine radiofrequency (RF) coil was used. The following scans were collected of the brain: 1) High-resolution 3D fast-field echo (FFE) T1-weighted sequence (repetition time (TR) = 9.9 ms, echo time (TE) = 4.6 ms, matrix = 232×192 , in-plane field of view (FOV) = 232×232 mm, slice thickness = 1 mm, number of slices = 150, number of signal averages = 1, flip angle = 8° , turbo field echo (TFE) factor = 192, scan time = 5 min 20 s) and 2) 32-echo gradient and spin echo (GRASE) T2-weighted sequence (TR = 1200 ms, TE = 10, 20, 30, ..., 320 ms, flip angle = 90° , refocusing flip angle = 180° , matrix = 224×90 , in-plane FOV = 224×180 mm, slice thickness = 1.5 mm, number of slices = 80, number of signal averages = 1, scan time = 9 min 30 s).

To acquire images of the cervical spinal cord (C2-C4), the following scans were collected: 1) merged fast-field echo (M-FFE) T1-weighted sequence (TR = 650 ms, first TE = 4.1 ms, number of echoes = 4, echo spacing = 7 ms, matrix = 188×188 , in-plane FOV = 150×150 mm, slice thickness = 2.5 mm, number of slices = 16, number of signal averages = 2, flip angle = 28° , scan time = 4 min 10 s) and 2) 32-echo GRASE T2-weighted sequence (TR = 1500 ms, TE = $n \times 10$ ms, flip angle = 90° , refocusing flip angle = 180° , matrix = 240×196 , in-plane FOV = 180×147 mm, slice thickness = 2.5 mm, number of slices = 16, number of signal averages = 1, scan time = 9 min 10 s).

For individuals with SCI, MRI-acquisition occurred within 6 weeks prior to downhill training (range = 3 to 39 days) and within 24 h of the final training session. For uninjured control participants, two sets of data were collected within a single session. Scans of the brain and spinal cord

were performed back-to-back, separated by changing of RF coils.

2.5. Generation of MWF maps

Generation of brain and spinal cord MWF maps from T2 relaxation imaging used the latest non-negative least squares method (Whittall et al., 1997), implemented with an extended phase graph algorithm to correct for stimulated echoes (Prasloski et al., 2012). The T2 distribution of each imaging voxel was partitioned into short (<40 ms), intermediate (40–200 ms), and long (>1500 ms) components using an in-house MATLAB code (MATLAB R2013b, MathWorks Inc) developed at the University of British Columbia (UBC; code requests: <https://mriresearch.med.ubc.ca/news-projects/myelin-water-fraction/>). The MWF is the sum of the amplitudes within the short T2 component divided by the amplitude sum of the total T2 distribution. Voxel-based MWF maps were generated for each participant.

2.6. Processing of brain imaging

Further processing of brain MRI data was performed as previously described by Lakhani et al. (2017). Here, we utilized the freely available FreeSurfer Software Suite v5.3 (<https://surfer.nmr.mgh.harvard.edu/>) to perform all morphometric procedures, including automated cortical reconstruction and volumetric segmentation (technical details described previously (Dale et al., 1999; Deoni et al., 2015; Fischl and Dale, 2000; Fischl et al., 2001; Fischl et al., 2002; Fischl et al., 2004; Lakhani et al., 2017; Reuter et al., 2010; Reuter et al., 2012; Salat et al., 2009; Segonne et al., 2004)). Processing included motion correction and averaging (Reuter et al., 2010) of multiple volumetric T1-weighted images, removal of non-brain tissue using a hybrid watershed/surface deformation procedure (Segonne et al., 2004), automated Talairach transformation, segmentation of the subcortical white matter (Fischl et al., 2002; Fischl et al., 2004), intensity normalization (Sled et al., 1998), tessellation of the gray matter white matter boundary, automated topology correction (Fischl et al., 2001; Segonne et al., 2007), and surface deformation following intensity gradients to optimally place the gray/white and gray/cerebrospinal fluid borders at the location where the greatest shift in intensity defines the transition to the other tissue class (Dale et al., 1999; Dale and Sereno, 1993; Fischl and Dale, 2000). Images

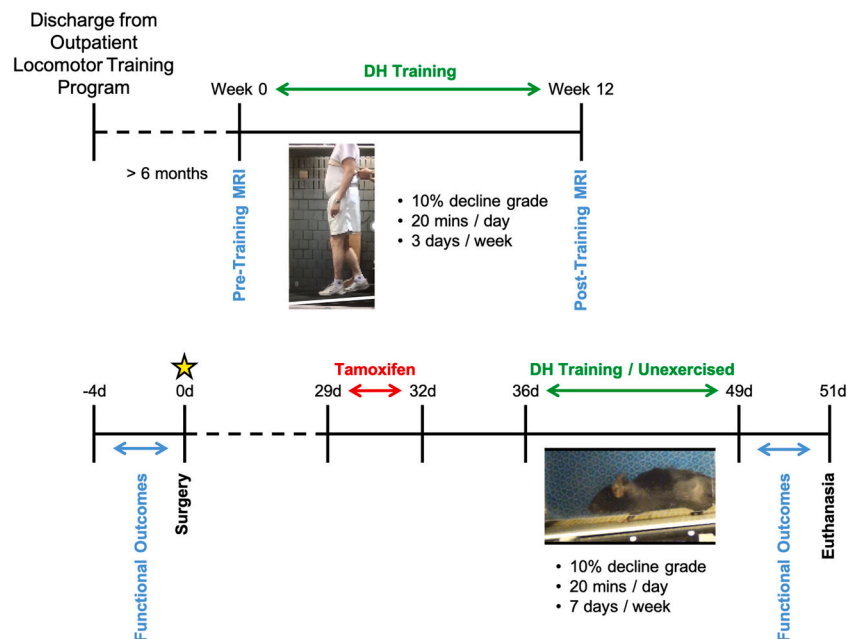


Fig. 1. Translational research design and experimental timelines. Individuals or mice with SCI completed the experiments described above. In mice, group randomization is denoted by a star. Abbreviations: DH = Downhill.

were automatically processed with the longitudinal stream (Reuter et al., 2012). Specifically, an unbiased within-subject template space and image was created using robust, inverse consistent registration (Reuter et al., 2010). Processing steps, such as skull stripping, Talairach transforms, atlas registration as well as spherical surface maps and parcellations were then initialized with common information from the within-subject template, significantly increasing reliability and statistical power (Reuter et al., 2012). Visual inspection of FreeSurfer surface reconstructions occurred at each stage of the longitudinal processing pipeline. Manual correction of errors included removing dura misidentified as cortex and re-processing the current and subsequent steps.

Linear co-registration occurred in three steps, using established methods by Lakhani et al. (2017) and Deoni et al. (2015): 1) registration of the first echo from the T2 scan to the MWF map, 2) registration of the first echo of the T2 scan to the parcellated T1 anatomy, and 3) registration of the MWF map to the parcellated anatomy. All processing occurred on the same workstation, using the same software version (Gronenschild et al., 2012), and individualized intensity normalization was used to increase registration accuracy (Sled et al., 1998). The FreeSurfer-generated white matter parcellation map (Salat et al., 2009) was used to extract mean MWF from *a priori* identified regions of interest (ROI).

2.7. Processing of spinal cord imaging

Spinal cord MRI data post-processing occurred as described by Liu et al. (2020). Spinal cord segmentation of the GRASE image occurred using the Spinal Cord Toolbox PropSeg tool (De Leener et al., 2017; Yiannakas et al., 2016), which generated a mask of the whole cord. Segmentation used the 16th echo of the GRASE image, because it has the greatest contrast between the spinal cord and cerebrospinal fluid. Visual inspections occurred and manual corrections to the mask were made when needed. The whole cord mask was then registered to the PAM50 template (De Leener et al., 2018) using a nonlinear transformation that involved a combination of center of mass rotation, affine, symmetric normalization, and bsplinesyn algorithms. This process obtained the warping functions used to transform the GRASE images and subsequently MWF maps into the template space. Visual inspection of all registered images occurred. MWF values were extracted from the PAM50 atlas labels.

2.8. MRI data analysis

For analysis of MWF data, subcortical white matter regions corresponding to areas known to be responsive to motor learning were identified *a priori*. Motor learning areas included white matter underlying the precentral, postcentral, caudal middle frontal, inferior parietal, and precuneus cortices, and the anterior and posterior regions of the corpus callosum (CC) (Lohse et al., 2014). Regions not expected to change following an eccentric-focused locomotor task were also identified *a priori*. Non-motor learning brain regions included white matter underlying the fusiform, inferior temporal, middle temporal, superior temporal, lateral orbitofrontal, lateral occipital, and lingual cortices (Lohse et al., 2014). We measured white matter plasticity in the cervical cord because frank demyelination, above-level atrophy, and neuroplasticity are localized here (Freund et al., 2012; Liu et al., 2017; Raineteau and Schwab, 2001). Due to the bilateral nature of the locomotor task, MWF data from right and left ROIs were combined and averaged to calculate an overall MWF value for each region.

We determined reliability of MWF measures by calculating intraclass correlation coefficients (ICC). Regions with poor reliability (ICC < 0.5) were excluded from subsequent analyses (Koo and Li, 2016). Next, we determined whether individual MWF changes surpassed measurement error (Beckerman et al., 2001) by calculating the minimal detectable change (MDC) for each ROI using the formula $MDC = SD * \sqrt{1 - r} * 1.96$ as described by Forrest et al. (2014). Here, SD represents the

standard deviation (SD) of the difference between pre- and post-intervention measurements, r represents the ICC, and 1.96 is the 97.5% quantile of the standard normal distribution of a two-tailed test with the significance set at $\alpha = 0.05$.

We compared the mean percent change of the downhill trained SCI participants (SCI + DH) with the mean percent change of the within-session data in the uninjured control group (Control) using a Mann Whitney U, nonparametric test to determine whether the mean percent MWF change within each ROI for the SCI + DH group exceeded the variability of the measure established by the Control group. Values are expressed as mean $\pm$ SD.

2.9. Experimental procedures in mice

Animal experiments were conducted with approval and in accordance with The Ohio State University Institutional Animal Care and Use Committee. Adult, male and female mice were group housed (3–5 per cage), fed standard chow *ad libitum*, and maintained on a 12-h light/dark cycle throughout the experiments.

2.10. Generation of PDGFR α -CreERT2:mT/mG (mT/mG) transgenic mice

Because OPCs express platelet-derived growth factor receptor α (PDGFR α), we utilized Cre-inducible PDGFR α mice (Rivers et al., 2008) to generate inducible reporter mice as described by Hesp et al. (Hesp et al., 2015). Heterozygous PDGFR α -CreER^{T2} mice were backcrossed with C57BL/6J mice for four generations, then bred with mice that express membrane-bound tdTomato on all cells prior to Cre-mediated recombination (B6.129(Cg)-Gt(ROSA)26Sor^{tm4}(ACTB-tdTomato,EGFP)^{Luo}/J; Jackson Laboratories) (Muzumdar et al., 2007), to generate PDGFR α -CreER^{T2}:mT/mG mice. Recombination was induced by oral tamoxifen administration (300 mg/kg dissolved in corn oil) *via* gavage for four consecutive days. During tamoxifen treatment, mice were given prophylactic subcutaneous injections of sterile saline (2 cc) and high calorie food supplements daily to prevent dehydration and weight loss. After recombination, cells transcribing the PDGFR α promoter down-regulate mT and begin to express membrane-bound green fluorescent protein (GFP), which is maintained for the life of the cell. Randomization into downhill training (SCI + DH) or unexercised control (SCI Unex) groups occurred immediately prior to injury. The disposition of all mTmG mice are described in Supplementary Fig. 2.

2.11. Experimental SCI procedures

Incomplete spinal cord contusion was performed (Faw et al., 2018). Mice were anesthetized *via* intraperitoneal ketamine (138 mg/kg) and xylazine (20 mg/kg). Prophylactic antibiotics (gentocin, 1 mg/kg) were administered subcutaneously prior to surgery. The spinal cord was exposed by removal of the T9 lamina. The Infinite Horizon device delivered a 75 kilodyne (kdyn) contusive force to the spinal cord. Paraspinal muscles were sutured, and the skin was closed using surgical clips. Mice received sterile saline (2 cc) to prevent dehydration and recovered overnight in warm cages. Postsurgical care included daily subcutaneous injections of prophylactic gentocin and saline for 5 days and twice-daily manual bladder expression for the duration of the experiments. Animals with impactor force or displacement values that exceeded ± 2 SD from the overall mean were excluded from analyses.

2.12. Rehabilitation paradigm

Mice received eccentric-focused, downhill treadmill training from 36 to 49 days post-injury, which consisted of quadrupedal stepping on a custom-built treadmill (Columbus Instruments; Columbus, OH) at the same 10% decline grade used for humans. Animals performed two, 10-min bouts of downhill stepping separated by a 20-min rest break in order

to deliver 1000+ steps during each 20-min session (Cha et al., 2007). Training speed was initially set at a comfortable pace (7.0 m/min) and progressively increased within and between sessions to the fastest speed that did not degrade stepping performance. Training progression was individualized based on performance. Manual assistance was provided as needed for trunk stability during locomotion. For both mice and humans, training paradigm length (weeks) was set at 150% of previously established flat treadmill rehabilitation efficacy (Mice = 1 week; Humans = ~6 weeks) (Hansen et al., 2013; Morrison et al., 2018).

2.13. Open field locomotion

Locomotor recovery was assessed in the open field using the Basso Mouse Scale (BMS) (Basso et al., 2006) prior to SCI, at 1 and 7 days post-injury, and weekly thereafter. Two raters, blind to group assignment, rated locomotor performance over 4 min considering joint movement, weight support, plantar stepping, coordination, paw position, trunk and tail control. Scores range from no hindlimb movement (0) to normal locomotor function (9). To maximize consistency, screening of BMS scores occurred at 1 day post-injury and mice with 1 day BMS scores >0.5 were excluded (Hansen et al., 2013).

2.14. Two-dimensional kinematics

Fine characteristics of locomotion were assessed prior to SCI and 49 days post-injury using two-dimensional kinematics during flat treadmill walking as described previously (Hansen et al., 2013). Briefly, animals were anesthetized with 2% isoflurane, the left hindlimb was shaved, and bony prominences (iliac crest and greater trochanter) were identified by palpation and passive range of motion. Depilatory cream removed hair from bony prominences to maintain marker locations for the duration of the study. The lateral malleolus and fifth metatarsal head were marked with white paint immediately prior to assessment. We recorded 20 steady-state hindlimb steps per animal per timepoint at 60 hertz using a Nikon D3200 digital video recorder. Movies were uploaded and markers were digitized using PEAK Motus software. Due to movement of the knee joint under the skin during locomotion, a triangulation program estimated the knee position based on post-mortem femur and tibia measurements (Goslow Jr. et al., 1973; Hansen et al., 2013). Positional data (X,Y coordinates) were obtained at distinct gait phases: Initial Contact (E1; end of swing), Lift Off (E3; end of stance), and Flexion (F; mid-swing). End point kinematic data were compared between SCI + DH and SCI Unex groups using independent samples *t*-tests.

2.15. Thermal sensitivity

Sensitivity to thermal stimuli was assessed at a consistent time of day at baseline and at six weeks post-injury using Hargreaves' Method (Hargreaves et al., 1988) (Plantar Test Apparatus; Ugo Basile, Comerio, Italy). A male rater, blind to experimental groups, performed all testing, which eliminated sex effects of the rater (Sorge et al., 2014). Mice acclimated for 30 min prior to testing in individual clear plastic compartments (11 cm × 17 cm) with a glass floor. Infrared, radiant heat was applied to the plantar L5 dermatome through the glass until a withdrawal response occurred. A photocell on the heat source determined withdrawal and the duration (in seconds) of heat application (latency to withdraw). To prevent tissue damage, a 20 s maximum application was set. Mice were tested randomly with 5 trials per hind paw. The longest and shortest latencies were dropped for each paw before averaging the remaining latencies to yield a single withdrawal threshold for each mouse. End point latencies were divided by pre-injury latencies and scores are expressed as a percent of baseline. An independent samples *t*-test was used to compare SCI + DH and SCI Unex groups.

2.16. Tissue processing

Mice were deeply anesthetized with ketamine (207 mg/kg) and xylazine (30 mg/kg) and transcardially perfused with 0.1 M PBS followed by 4% paraformaldehyde (PFA) in PBS. Spinal cords were removed, including the cervical enlargement, divided into C1-C3 and C4-T1 segments, and a 6 mm thoracic segment centered on the injury site (SCI groups) or T9 (Naive). Tissue was post-fixed in 4% PFA for 1 h at 4 °C, placed in 0.2 M phosphate buffer overnight, then cryoprotected in 30% sucrose, before being embedded in optimal cutting temperature compound (ThermoScientific) and frozen on dry ice. Upper cervical and thoracic segments were oriented for cutting in the transverse plane. Lower cervical segments were oriented in the coronal plane to allow analysis along the length of axons. All tissue was cut at 20 µm using a Microm HM 505E cryostat and stored at -80 °C.

2.17. Immunofluorescence

To label the lesion border, axons, or paranodes, slides were thawed and rehydrated in 0.1 M PBS, blocked with 4% bovine serum albumin with 0.3% Triton X-100 in PBS, incubated in primary antibody overnight (Supplementary Table 2), then in secondary antibody (Supplementary Table 2) for 40–60 min before rinsed in PBS and coverslipped with ProLong Diamond Antifade Mountant with DAPI (ThermoFisher).

To identify new oligodendrocytes, slides were thawed and rehydrated in 0.05 M tris buffered saline (TBS), blocked with 10% fetal bovine serum with 0.5% Triton X-100 in TBS, incubated in primary antibody overnight (Supplementary Table 2), then in secondary antibodies (Supplementary Table 2) at room temperature for 60–90 min, rinsed with TBS then PBS before being coverslipped as above.

2.18. Digital image analysis

Fluorescent images were captured by confocal microscopy (Olympus FV1000 laser scanning confocal microscope; OSU Campus Microscopy and Imaging Facility). All images were blinded prior to quantification, which was performed by a single rater. Secondary antibodies at 633 or 647 wavelength were pseudocolored magenta during microscopy.

To quantify epicenter lesion area, spinal cord images stained for glial fibrillary acidic protein (GFAP) were imaged at 10× magnification and analyzed using stereological point counting in Stereo Investigator (MBF Bioscience). Using the Cavalieri method, a point grid was placed randomly over the epicenter section and intersections over tissue were identified as either spared or lesioned tissue, representing a point (p). Point counts for each section were converted to area estimates using the formula: Estimated Area = $(\sum P) \cdot (a/p)$, where $\sum P$ is the sum of points for an individual mouse and a/p is the area of each point. We calculated the percent sparing by dividing the area of GFAP⁺ tissue by the total cross-sectional area. Animals whose percent sparing exceeded ±2 SD from the mean were excluded from the study. Independent samples *t*-test compared percent sparing between SCI + DH and SCI Unex groups.

To determine the density and location of new oligodendrocytes in the cervical spinal cord, CC1⁺ cells that fit the proper size and shape criteria for oligodendrocytes (~10 µm, round/oval), and colocalized with GFP and DAPI (Hesp et al., 2015; McKenzie et al., 2014) were quantified throughout spinal cord hemisections located at C2 or C3. Spinal white matter was subdivided into dorsal columns, lateral, and ventral funiculi. To delineate lateral and ventral funiculi, a line was drawn from the midpoint of the ventral horn gray matter to the anterolateral sulcus at the edge of the tissue (Chung and Coggeshall, 1983). Cell counts were divided by the funiculi area, determined using ImageJ (<https://imagej.nih.gov/>), and are expressed as the number of CC1⁺/GFP⁺/DAPI⁺ new oligodendrocytes per square millimeter (mm²). One-way ANOVA with Tukey's *post hoc* test analyzed CC1/GFP/DAPI⁺ cell density between Naive, SCI + DH, and SCI Unex groups.

To quantify oligodendrocyte lineage cell and axon interactions,

GFP⁺/neurofilament (NF)⁺ axon segments were quantified from high magnification (60 $\times$) images in the gracile tract of horizontally cut, cervical cord (C4-T1) tissue sections. Axon contacts were counted if GFP labeling surrounded NF⁺ axons (in the same plane of focus) over at least a 5 μ m distance. Quantifications are expressed as the number of GFP⁺/NF⁺ axon segments per mm² and analyzed between Naive, SCI + DH, and SCI Unex groups using a one-way ANOVA with Tukey's *post hoc* test. Representative optical z-stacks were acquired at high magnification (180 $\times$) to illustrate GFP⁺ processes wrapping NF⁺ axons.

Contactin-associated protein (Caspr)⁺ paranodes in the gracile tract of the cervical enlargement (C4-T1) were quantified as a surrogate measure of functional myelin. High magnification (80 $\times$) images were batch processed using MIPAR Image Analysis Software (Worthington, OH). Briefly, a threshold of positive labeling was determined for Caspr⁺ labeling. Caspr⁺ profiles were counted and binned into short (0.5–1.99 μ m) or long (2.0–19.0 μ m) profiles. The number of short and long profiles are expressed as a percent of total Caspr⁺ profiles. Between-group analysis was performed using one-way ANOVA with Tukey's *post hoc* test.

2.19. Statistics

Data from human participants are expressed as either individual values or mean $\pm$ SD. For within-subject comparisons, individual differences between pre- and post-training MWF values were compared to the calculated MDC for brain and spinal cord ROIs. Levene's test showed there was not a statistically significant difference in variances between SCI + DH and Control groups. As such, we proceeded with between-group comparisons using a Mann-Whitney U, nonparametric test. For animal experiments, values are expressed as mean $\pm$ standard error of the mean (SEM). Individual data points >2 SD above or below the mean were considered outliers and excluded from subsequent analyses. Between-group comparisons were made using either independent samples t-test or one-way ANOVA with Tukey's *post hoc* testing. Relationships between functional and immunohistochemical measures were determined using Pearson's correlation. All comparisons occurred using GraphPad Prism statistical software. Significance was set at $\alpha \leq 0.05$ for all statistical tests.

2.20. Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

3. Results

3.1. Downhill training promotes white matter plasticity in brain motor learning regions and spinal cord ventral funiculi in chronic, incomplete human SCI

We employed a translational paradigm to determine whether downhill training promoted white matter plasticity in chronic SCI (Fig. 1). In human participants, we first determined the stability of MWF values at our OSU facility by measuring the within-session test-retest reliability of the MWF signal in uninjured control participants (Supplementary Table 3). Missing temporal lobe imaging data was identified on the GRASE scan of three uninjured control individuals that affected the fusiform, inferior temporal, middle temporal, and superior temporal regions. Thus, comparisons occurred using data from $n = 4$ uninjured control participants without data loss for these regions. Two motor learning regions (paracentral and superior frontal) had poor single-session test-retest reliability and were excluded from further analyses. Moderate to excellent reliability of MWF occurred for remaining ROIs (ICC range: 0.503–0.907; Supplementary Table 3). Spinal cord ROIs had good to excellent test-retest reliability of MWF values (ICC range: 0.814–0.917; Supplementary Table 3).

To determine whether individual, pre-post changes in MWF were likely to exceed measurement error (Forrest et al., 2014), we calculated the MDC. In brain regions, MDCs ranged from 0.0032 in the middle temporal region to 0.0122 in the posterior corpus callosum (Supplementary Table 4). Greater MDC variability occurred in spinal cord regions, ranging from 0.0028 in the whole cord to 0.0389 in the lateral funiculi (Supplementary Table 4).

Individual responses to downhill training were determined by comparing pre-training MWF values in motor learning and non-motor brain regions with those post-training (Fig. 2A–C). Participant SCI-01 had greater MWF after downhill training that exceeded the MDC in 6 out of 7 motor learning regions, with several regional increases being substantial and the smallest remaining above 10% (Fig. 2D). Substantial increases of 26% for sensory, 33% for motor and 40% in caudal middle frontal regions were observed. Conversely, 1 non-motor learning region showed an increase and 1 showed a decrease in MWF that exceeded the MDC after training (Fig. 2D). In the cervical spinal cord, MWF was decreased in the dorsal columns but increased in the ventral funiculi after downhill training (Fig. 2G).

Participant SCI-02 had MWF increases exceeding the MDC across 6 out of 7 brain motor learning regions after downhill training (Fig. 2E). The largest increase occurred in the anterior corpus callosum (24%), followed by precuneus (22%) and posterior corpus callosum (20%). In non-motor learning regions for SCI-02, MWF increased (lingual region) and decreased (lateral orbitofrontal region) beyond MDC. No spinal cord MWF changes exceeded the MDC (Fig. 2H).

Participant SCI-03 showed the fewest brain changes in response to downhill training (Fig. 2F). For this participant, MWF increased in 2 of 7 motor learning regions and decreased in 1 of 7 non-motor learning regions relative to MDC after training. The greatest increase in MWF occurred in the posterior corpus callosum (12%), followed by post-central (10%). A MWF increase exceeding the MDC occurred in the cervical ventral funiculus after downhill training (Fig. 2I).

Next, we compared the average percent change between pre- and post-training values in the downhill trained individuals with SCI (SCI + DH) to within-session percent change values in uninjured control participants (Control; Fig. 3A–T). In the brain, downhill training significantly increased MWF in 2 of 7 motor learning regions (Fig. 3E; postcentral, $P = 0.0167$, 95% CI = -33.46 to -4.75 ; and Fig. 3G; precuneus, $P = 0.0167$, 95% CI = -34.40 to -5.49) and combined percent change across all motor learning regions (Fig. 3H; $P = 0.033$, 95% CI = -24.49 to -0.28). No between-group differences in MWF occurred in any individual region or when combined across all non-motor learning regions (Fig. 3I–P; $P > 0.05$). In the cervical spinal cord, ventral funiculi MWF significantly increased after downhill training compared to the control group (Fig. 3S; $P = 0.033$, 95% CI = -31.66 to -1.16).

3.2. Downhill training promotes oligodendrogenesis rostral to SCI

The mechanisms underlying training-induced white matter plasticity after CNS damage are poorly understood. To begin to address this knowledge gap, we conducted a downhill rehabilitation experiment in mice with chronic, incomplete SCI like that performed in human participants. Mice were either uninjured (Naive), injured without training (SCI Unex), or injured and downhill trained (SCI + DH). There were no between-group differences in SCI severity metrics [impactor force (SCI + DH = 77.0 ± 0.7 kdyn, SCI Unex = 77.0 ± 0.8 kdyn; $P > 0.05$), displacement (SCI + DH = 599.2 ± 32.6 μ m, SCI Unex = 561.0 ± 13.3 μ m; $P > 0.05$), lesion area (SCI + DH = $45.6 \pm 2.5\%$, SCI Unex = $47.7 \pm 2.1\%$; $P > 0.05$)].

After SCI, downhill training increased GFP⁺/CC1⁺ new oligodendrocytes in the cervical dorsal columns (Fig. 4A–B; $P = 0.006$ vs. Naive, $P = 0.033$ vs. SCI Unex) and overall white matter (Fig. 4D; $P = 0.027$ vs. Naive). Chronic oligodendrogenesis occurred in the cervical lateral white matter regardless of downhill training but was less consistent in untrained animals (Fig. 4C; SCI + DH, $P = 0.015$ vs. Naive; SCI Unex, P

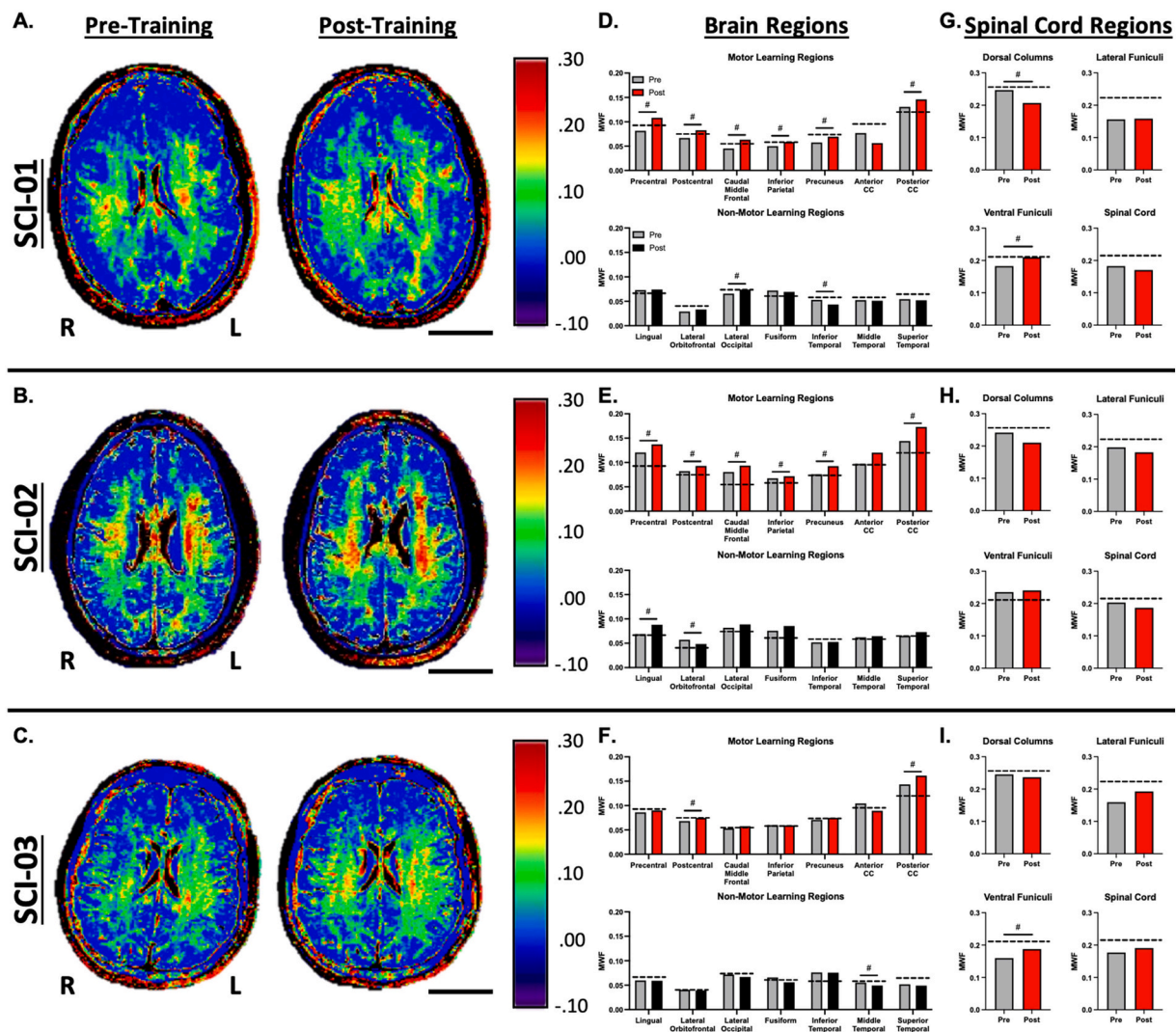


Fig. 2. Downhill training increases brain and spinal cord MWF in individuals with chronic, incomplete SCI. Pre- and post-downhill training MWF maps are shown from three individuals with SCI (A–C). Scale bar = 5 cm. Quantifications of MWF within brain (D–F) and spinal cord (G–I) regions of interest are shown. Dashed lines represent average MWF of Control participants. # = individual change exceeded the MDC for that region. Abbreviations: CC = Corpus Callosum; DH = Downhill; MDC = Minimal Detectable Change; MWF = Myelin Water Fraction; SCI = Spinal Cord Injury.

= 0.042 vs. Naive). There was no difference in ventral white matter new oligodendrocytes (Naive = 0.012 ± 0.002 cells per mm^2 , SCI + DH = 0.005 ± 0.002 cells per mm^2 , SCI Unex = 0.007 ± 0.003 cells per mm^2 ; $P > 0.05$). Thus, downhill training in mice with chronic SCI stimulates oligodendrogenesis in the dorsal columns rostral to the injury site.

3.3. Downhill training increases oligodendrocyte lineage cell interactions with axons and normalizes nodes of ranvier

As expected, oligodendrocyte lineage cell processes contacted and wrapped a subset of NF+ axons in the dorsal columns (Fig. 5A and B). Despite an overall loss of axons, SCI increased the number of OPC-axon contacts 4-fold (Fig. 5C; SCI Unex, $P = 0.044$ vs. Naive). Downhill training potentiated the increased number of OPC-axon contacts, which increased by 11-fold vs. Naive (Fig. 5C; $P < 0.0001$ vs. Naive and SCI Unex). Myelin pathology disrupts nodal structure, which can be evidenced by altered Caspr⁺ profile length. SCI caused a significant shift to shorter Caspr⁺ profile length compared to Naive (Fig. 6B and C, $P = 0.008$). Downhill training produced an intermediate distribution of profile lengths that was not significantly different from Naive.

3.4. Downhill training normalizes body position during locomotion in SCI mice

Functionally, downhill training normalized the aberrant, hypermetric body position displayed by untrained SCI mice during flat treadmill walking (Fig. 7A). Downhill-trained mice had significantly reduced hip height (Fig. 7B–D; Initial Contact, $P = 0.0004$; Lift Off, $P = 0.015$; Flexion, $P = 0.0005$) and knee angle (Fig. 7H and J; Initial Contact, $P = 0.004$; Flexion, $P = 0.04$) compared to untrained SCI animals. Lower hip height at the end of stance (Lift Off), when eccentric activity is expected to be highest, was significantly correlated with increased new oligodendrocytes in the dorsal columns (Fig. 7F). There were no differences in gross open field locomotion as measured with BMS (SCI + DH = 6.2 ± 0.5 , SCI Unex = 6.3 ± 0.4 ; $P > 0.05$).

3.5. Downhill training normalizes sensory function in chronic SCI

Downhill training also improved below-level sensory function in SCI mice. Untrained mice had significantly faster hindlimb withdrawal to thermal stimulus than downhill-trained mice (Fig. 8A; $P = 0.03$), who displayed no hyperalgesia, with higher thresholds approximating baseline values. A more normal, longer withdrawal latency was significantly

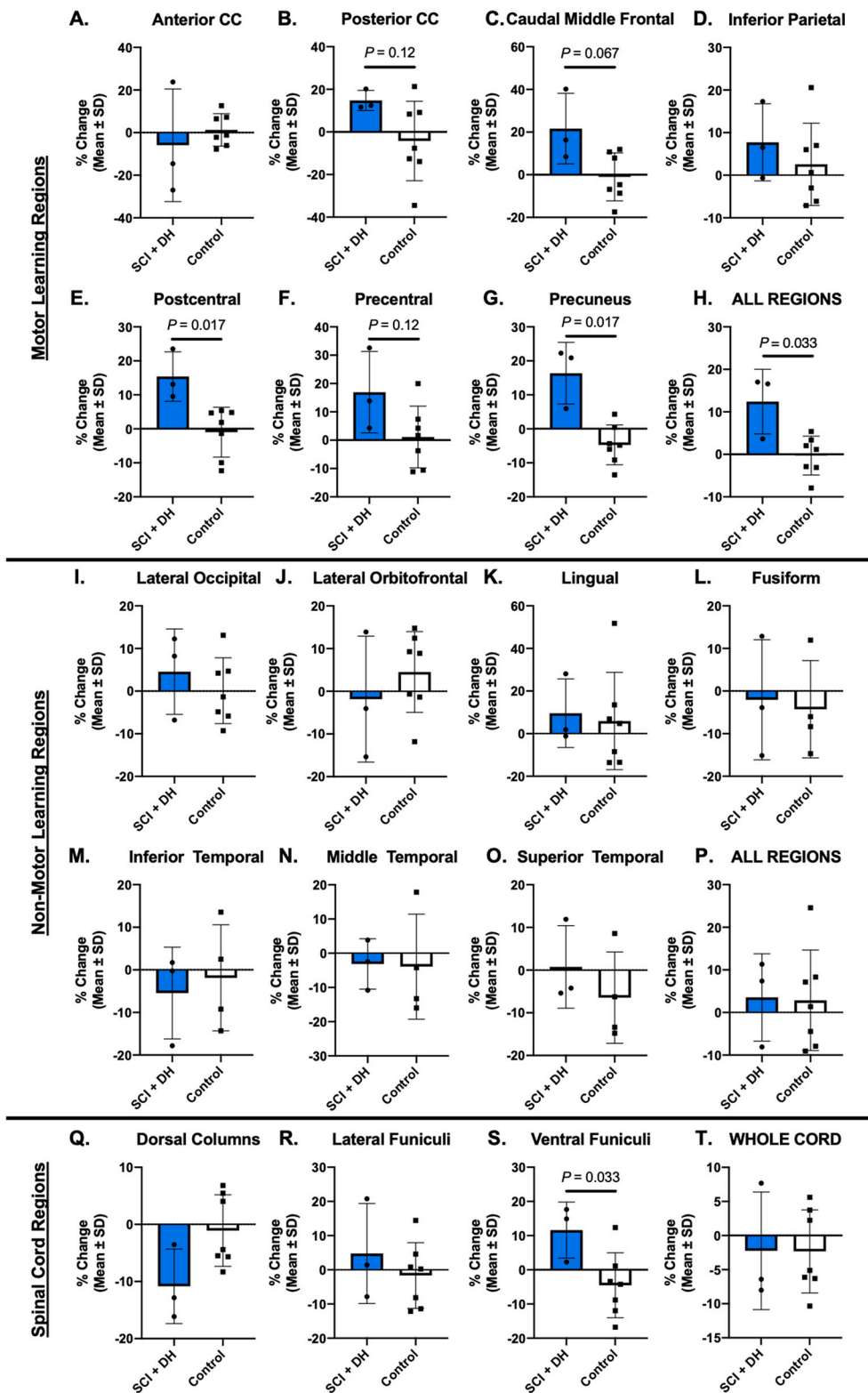


Fig. 3. Downhill training-induced MWF increases in humans with chronic, incomplete SCI exceed measurement variability of control participants. Between-group comparison of percent MWF change by region of interest associated with brain motor learning and non-motor learning areas, spinal cord white matter regions and whole spinal cord at C2. P -values are shown. Mann-Whitney U Test. Abbreviations: CC = Corpus Callosum; DH = Downhill; MWF = Myelin Water Fraction; SCI = Spinal Cord Injury.

correlated with an increased density of new oligodendrocytes in the dorsal columns (Fig. 8B; $r = 0.682$, $P = 0.026$) but not lateral funiculi (Fig. 8C; $r = 0.542$, $P = 0.085$).

4. Discussion

There is mounting clinical and preclinical evidence that myelin

plasticity accompanies motor skill learning in the healthy CNS (Hofstetter et al., 2013; Lakhani et al., 2016; McKenzie et al., 2014; Sampaio-Baptista et al., 2013; Scholz et al., 2009). Recent evidence in uninjured mice suggests that oligodendrogenesis and adaptive myelination are required for complex motor skill learning (Marques et al., 2016; McKenzie et al., 2014; Xiao et al., 2016). Here, we provide the first evidence of training-induced white matter plasticity in humans and

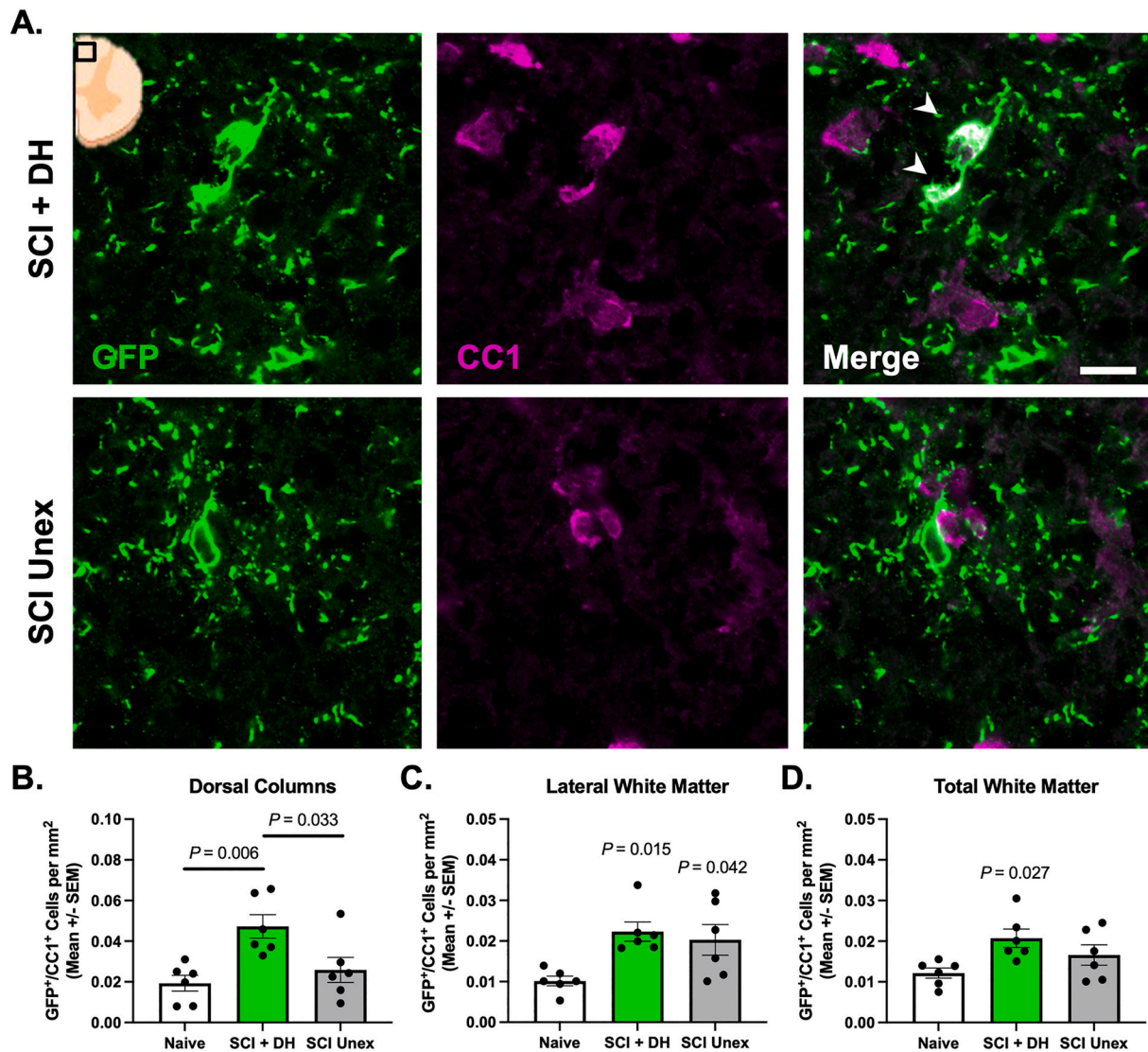


Fig. 4. Downhill training promotes oligodendrogenesis rostral to SCI in mice. (A) Single channel and merged confocal images of oligodendrocyte lineage cells (GFP⁺; green) and CC1⁺ oligodendrocytes (magenta) from cervical cord dorsal column white matter. GFP⁺/CC1⁺ new oligodendrocytes (white arrows) significantly increased with downhill training in the dorsal columns (B), and lateral white matter (C). A trend occurred for total white matter (D). *P*-values are shown. One-way ANOVA with Tukey's *Post Hoc*. Scale Bar = 10 μ m. Abbreviations: DH = Downhill; GFP = Green Fluorescent Protein; SCI = Spinal Cord Injury; Unex = Unexercised. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

rodents with SCI and reveal a novel rehabilitation approach to promote oligodendrogenesis and functional recovery in the chronic phase of injury. Together, our studies suggest that eccentric-focused downhill training increased MWF in humans, likely through oligodendrogenesis and greater axon contacts consistent with new myelin in regions associated with training. In SCI mice, eccentric treadmill training induced white matter plasticity, normalized locomotor kinematics, and reduced neuropathic pain-like behavior.

Motor learning appears to rely on oligodendrogenesis and adaptive myelination via OPC differentiation in response to neurotransmitters released from axons repetitively activated by the training paradigm (McKenzie et al., 2014). Here, downhill eccentric training activated glutamatergic muscle spindle afferents in the dorsal columns of the spinal cord (Burke et al., 1978; Hulliger et al., 1985; Lay, 2005; Lay et al., 2006; Lay et al., 2007; Minetti et al., 2002). We theorize that the resulting differentiation of OPCs into myelinating oligodendrocytes in the gracilis tract may be a new activity-dependent mechanism for functional recovery (Bergles et al., 2000; Borges et al., 1994; Etxeberria et al., 2016; Gautier et al., 2015; Hamilton et al., 2009; Kukley et al.,

2007; Wake et al., 2011; Yamazaki et al., 2007; Ziskin et al., 2007). That thalamic neurons directly synapse on OPCs throughout the brain, including the premotor cortex and primary somatosensory cortex (Mount et al., 2019), suggests that brain and spinal cord regions may be primed to respond to training. Indeed, our finding that higher order skill learning centers in the brain had increased MWF after eccentric training in human SCI provides further support (Fang et al., 2001, 2004; Kwon and Park, 2011; Olsson et al., 2012; Perrey, 2018; Yao et al., 2014). In both clinical and preclinical SCI, eccentrically-driven myelin responses occurred in ascending and descending proprioception-related tracts (e.g., gracile, spino-olivary, and motor). Interestingly, lateral funiculi oligodendrogenesis was highly variable in the absence of training, occurring in some animals but not others. It was also not associated with any of the observed functional improvements. This raises questions as to the task specificity of the lateral funiculus response and whether downhill training may be a means to promote oligodendrogenesis and recovery in animals with a lesser spontaneous response to SCI. Future studies are required to address these important questions.

The role of myelin in promoting motor recovery after experimental

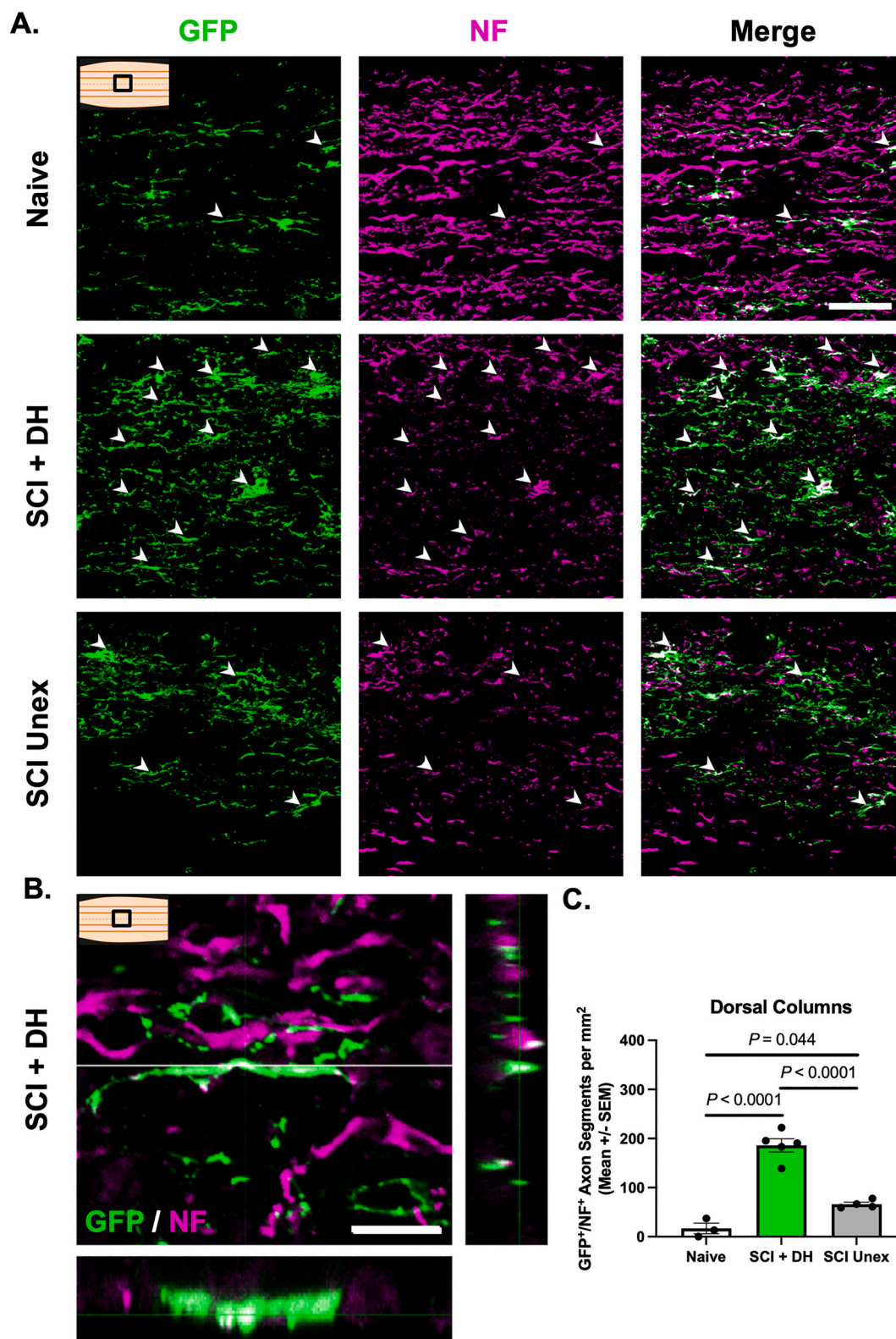


Fig. 5. Downhill training increases axon contacts by oligodendrocyte lineage cells in the gracile tract of the cervical cord. (A) Single channel and merged confocal images show oligodendrocyte lineage cell (GFP; green) and NF⁺ axon (magenta) contacts (white arrows). Scale bar = 50 μ m. (B) A GFP⁺ process (green) is shown wrapping a NF⁺ axon (white) consistent with new myelin. Spinal cord schematic in A and B shows the location of images within cervical cord. Scale bar = 10 μ m. (C) Quantification of GFP⁺/NF⁺ axon contacts is shown with corresponding *P*-values. One-way ANOVA with Tukey's *Post Hoc*. Abbreviations: DH = Downhill; GFP = Green Fluorescent Protein; NF = Neurofilament; Unex = Unexercised. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

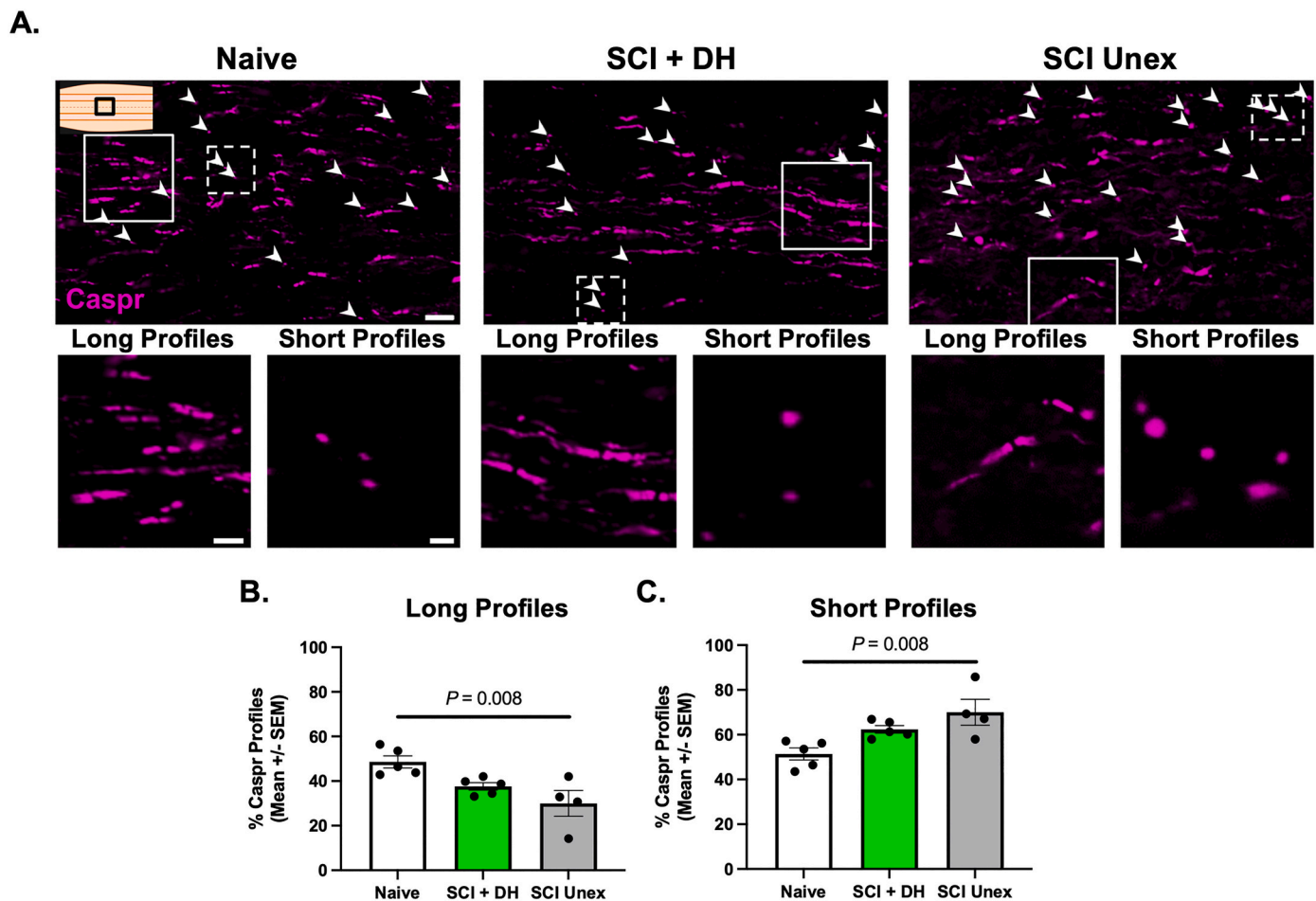


Fig. 6. Downhill training normalizes dorsal column paranodal structure in chronic SCI. (A) Confocal images show Caspr⁺ profiles in the cervical dorsal columns. Profiles are classified into long (2–19 μ m) and short (white arrows, 0.5–1.99 μ m). Scale bar = 10 μ m. Insets show long (solid) and short (dashed) profiles at high magnification. Scale bar = 2 μ m. (B, C) There was a significant shift toward short profiles in unexercised compared to Naive mice. Downhill training produced an intermediate phenotype that was not significantly different from Naive. *P*-values are shown. One-way ANOVA with Tukey's *Post Hoc*. Abbreviations: Caspr = Contactin Associated Protein; DH = Downhill; SCI = Spinal Cord Injury; Unex = Unexercised.

SCI is unclear because locomotor recovery can occur in the absence of remyelination by new oligodendrocytes (Duncan et al., 2018). Whether experimental conditions account for this spontaneous recovery warrant consideration and further study (Pukos et al., 2019). Regardless, rehabilitation appears to be a potent inducer of white matter plasticity in clinical and preclinical SCI. These findings align with motor learning and skill refinement via adaptive myelination in uninjured conditions (Lakhani et al., 2016; McKenzie et al., 2014; Sampaio-Baptista et al., 2013). Indeed, quantitative gains in locomotion occurred with eccentric training even in chronic SCI, improving hip and knee motor control in mice and postural balance and support in human SCI, which enabled one participant to use a cane instead of a walker (Supplementary Table 1). These improvements appear to increase efficiency of gait, thereby reducing fatigue and ultimately may mean the difference between home and community access.

The myelination and recovery response to eccentric downhill training likely results from accentuating the sensory-motor loop. Repetitive eccentric stepping activates muscle spindle afferents (Burke et al., 1978; Hulliger et al., 1985; Wilkinson et al., 2012), which modulate cord stretch reflexes so locomotion adapts to demands of the task (Akay et al., 2014; Beres-Jones and Harkema, 2004; Edgerton and Roy, 2009; Grillner and Zangger, 1984; Harkema et al., 1997; McCrea, 2001; Pearson, 2004; Rossignol et al., 2006; Timoszyk et al., 2005). In this way, precise integration of afferent sensory and descending motor signals produce controlled muscle lengthening, allowing power

absorption (Enoka, 1996). Indeed, downhill walking decreases spinal reflex excitability in healthy individuals (Hoque et al., 2018a) and those with multiple sclerosis (Hoque et al., 2019; Hoque et al., 2018b). Supraspinal processing of eccentrically-derived ascending muscle spindle signals regulate spinal cord motor neuron output required by complex tasks.

Rehabilitation timing is important. Effective interventions delivered too soon, in the face of high levels of inflammation, can be detrimental and yet are less effective when delivered late after SCI (Hansen et al., 2013). From an experimental design perspective, initiating downhill rehabilitation in the chronic phase of injury, when functional recovery has plateaued, increases the likelihood that any observed effects are related to the intervention rather than spontaneous recovery. This also sets the highest challenge for intervention efficacy, as many natural plasticity responses that are upregulated early after injury have returned to baseline (Fawcett, 2009). Strikingly, downhill training improved locomotion at a time when flat treadmill training is ineffective in mice (Hansen et al., 2013). McKenzie et al. (2014) showed that novel, complex training rather than simple or previously learned tasks promoted oligodendrogenesis and motor learning. In our studies, downhill eccentric walking required complex neuromotor control and was novel for both mice and humans - neither participated in downslope activities prior to training. Moreover, that white matter plasticity in the form of oligodendrogenesis (mouse) and increased MWF (human) occurred in response to chronic-delivered eccentric rehabilitation

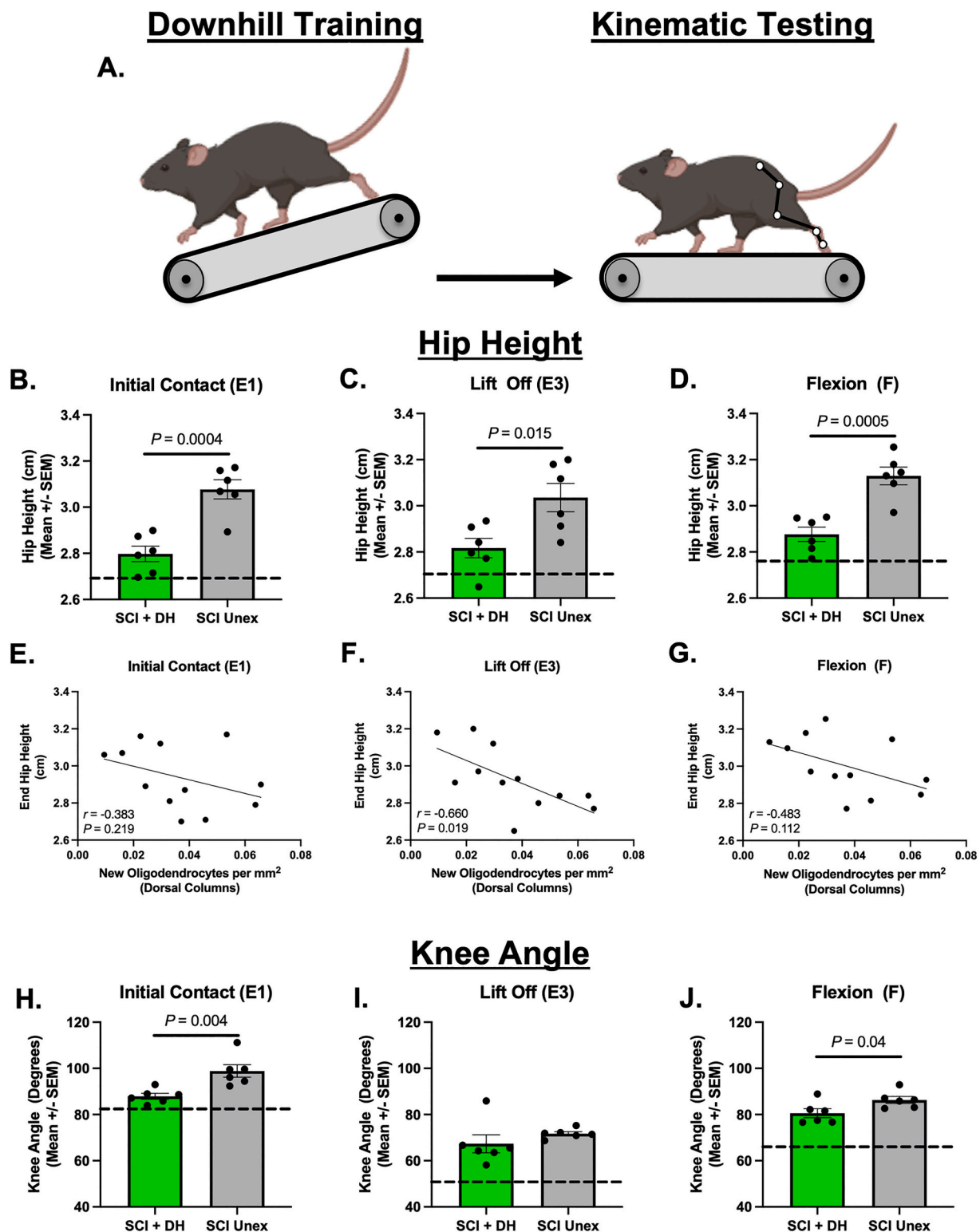


Fig. 7. Downhill training normalizes body position that is associated with dorsal column white matter plasticity. (A) Schematic shows downhill treadmill training and flat treadmill kinematic testing conditions. (B–D) Quantification of hip height at the end of distinct gait phases: (B) Initial Contact (E1), (C) Lift Off (E3), (D) Flexion (F). Dashed lines represent normal pre-injury performance. (E–G) A significant negative correlation was found between hip height and dorsal column new oligodendrocytes at Lift Off but not Initial Contact or Flexion. (H–J) Quantification of knee angle at the end of distinct gait phases. Higher degree values indicate greater knee extension. *P*-values are reported above bars (Independent Samples *t*-Test). Pearson's correlation *r* values are reported along with corresponding *P*-values. Abbreviations: DH = Downhill; SCI = Spinal Cord Injury; Unex = Unexercised.

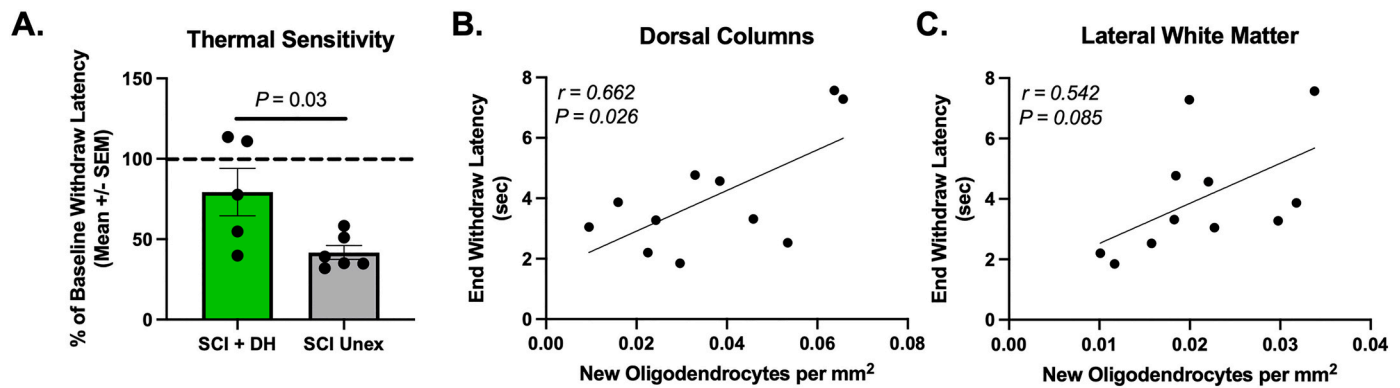


Fig. 8. Downhill training-induced reduction in hyperalgesia is associated with dorsal column white matter plasticity. (A) Quantification of below-level thermal heat sensitivity shows significantly normalized hindpaw withdraw latency in SCI + DH mice (Independent Samples *t*-Test). Dashed line represents pre-injury performance. Withdrawal latency at 49 days was positively correlated with new oligodendrocytes in the dorsal columns (B) but not the lateral white matter (C). Pearson's correlation *r* values are reported with corresponding *P*-values. Abbreviations: DH = Downhill; GFP = Green Fluorescent Protein; SCI = Spinal Cord Injury; Unex = Unexercised.

suggests that eccentric motor control is not learned during spontaneous recovery, self-rehabilitation, or locomotor rehabilitation on a flat treadmill after SCI. It is likely that initiating downhill rehabilitation in the subacute phase (2–4 weeks post-injury), when spontaneous oligodendrogenesis and remyelination are most robust (Hesp et al., 2015), will synergize with the natural recovery response to promote even larger gains in recovery than observed here. Future studies should investigate the optimal timing for downhill training to promote white matter plasticity and motor learning after SCI.

Eccentric downhill training reduced hyperalgesia-like sensation in mice with chronic SCI, extending our previous work that showed normalization of sensation and reduced pain-like behavior with flat treadmill training (Faw et al., 2018). Interestingly, greater oligodendrogenesis occurred in the cervical dorsal columns, not in the anterolateral pain pathways. Perhaps greater dorsal white matter improved transmission of mechanosensory fibers to the thalamus and reduced pain consistent with the gate theory (Melzack and Wall, 1965). Further research is needed to confirm mechanisms of pain reduction in response to downhill training.

Our study is not without limitations. Spine hardware, claustrophobia, and motion artifact contributed to numerous individuals with SCI being excluded from the human experiment. However, the within-subject analyses and nonparametric statistical methods appropriate for small sample sizes used here, as well as estimation-based approaches that are available in our preprint (Faw et al., 2020) give us confidence our findings will generalize to larger groups. Next, consideration of MWF as a measure of adaptive myelination carries risk. Although MWF values agree well with histopathologically-identified myelin (Chen et al., 2017; Laule et al., 2006), it cannot differentiate between the mechanisms of myelin plasticity, such as *de novo* myelination by newly formed oligodendrocytes, new myelin internodes, increased internode length, increased myelin thickness. It is also susceptible to CNS iron content (Birkl et al., 2019). While a combination of these mechanisms is likely involved in training-induced myelin plasticity and functional recovery, our preclinical data suggest that new myelin produced by newly differentiated oligodendrocytes plays a role. In preclinical SCI, new oligodendrocytes appear around the epicenter 2–3 days post-injury (Tripathi and McTigue, 2007), yet new myelin is delayed 2–3 weeks (Assinck et al., 2017; Duncan et al., 2018; Gledhill et al., 1973; Harrison and McDonald, 1977; Hesp et al., 2015; Lasiene et al., 2008; Powers et al., 2013). Similarly, motor skill training promotes new oligodendrocytes within days (Marques et al., 2016; McKenzie et al., 2014; Xiao et al., 2016), and unless newly formed oligodendrocytes undergo neural integration and wrapping in a short timeframe, they will apoptose (Barres et al., 1992; Hughes et al., 2018; Trapp et al., 1997). Myelin from

new oligodendrocytes is characteristically thin and progressively remodels over 6 months or more (Powers et al., 2013; Young et al., 2013). Thus, myelin plasticity after just two weeks of downhill training in the mouse likely reflects the leading edge of adaptive myelination. Greater myelin responses to rehabilitation may exist at later time points and hold even more potential for SCI recovery. Importantly, the proof-of-principle design used here compared downhill training to unexercised control groups (animal and human), which prevents differentiation between types of rehabilitation training to promote white matter plasticity and recovery in chronic SCI. Thus, comparisons between downhill training and either general exercise or flat treadmill training require future study.

Overall, these studies reveal an exciting relationship between white matter plasticity in training-induced sensorimotor recovery in preclinical and clinical SCI. Clinical studies provide the first evidence that novel, complex rehabilitation can induce greater white matter in motor learning regions of the brain. A well-controlled preclinical study identified white matter plasticity in sensorimotor regions and greater recovery of locomotion and sensation with downhill training. That these gains occurred chronically, when few effects were expected, confirms the potential of eccentric rehabilitation and supports further research into the dose, timing and molecular mechanisms of rehabilitation-induced white matter plasticity.

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Competing interests

The authors report no competing interests.

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